Statement of Business Problem

A leading retail company wants to better understand its customers' shopping behavior to improve sales, customer satisfaction, and long-term loyalty. There has been a change in buying behavior across demographics, product categories, and sales channels (online vs. offline) in the management team. They are especially interested in understanding what influences the consumer's decision and repurchase, such as discounts, reviews, seasons, or payment preferences.

Your job is to work with the company's consumer behavior data set and answer the following broad business question:

How can the company use consumer shopping data to detect patterns, improve consumer engagement, and refine marketing and product strategies?"

Deliverables

1. **Data Preparation & Cleaning (Python):**
Clean and prepare the raw dataset using Python. Handle missing values, remove duplicates, fix inconsistencies, and organize the data for analysis.
2. **Data Analysis (SQL):**
Use SQL queries to explore customer shopping behavior, identify customer segments, measure loyalty, and understand the factors influencing purchasing decisions.
3. **Data Visualization (Power BI):**
Build an interactive Power BI dashboard to visualize key metrics, shopping trends, and customer behavior, helping businesses make better decisions.
4. **Project Report & Presentation:**
Prepare a short report summarizing the main findings, business insights, and practical recommendations. Present the results in a clear, easy-to-understand format.
5. **GitHub Repository:**
Keep all Python scripts, SQL queries, Power BI dashboards, and project documentation neatly organized in a GitHub repository.